

Automated recognition and mapping of building management system (BMS) data points for building energy modeling (BEM)

Sicheng Zhan, Adrian Chong (✉), Bertrand Lasternas

Department of Building, School of Design and Environment, National University of Singapore, 4 Architecture Drive, Singapore 117566, Singapore

Abstract

With the advance of the internet of things and building management system (BMS) in modern buildings, there is an opportunity of using the data to extend the use of building energy modeling (BEM) beyond the design phase. Potential applications include retrofit analysis, measurement and verification, and operations and controls. However, while BMS is collecting a vast amount of operation data, different suppliers and sensor installers typically apply their own customized or even random non-uniform rules to define the metadata, i.e., the point tags. This results in a need to interpret and manually map any BMS data before using it for energy analysis. The mapping process is labor-intensive, error-prone, and requires comprehensive prior knowledge. Additionally, BMS metadata typically has considerable variety and limited context information, limiting the applicability of existing interpreting methods. In this paper, we proposed a text mining framework to facilitate interpreting and mapping BMS points to EnergyPlus variables. The framework is based on unsupervised density-based clustering (DBSCAN) and a novel fuzzy string matching algorithm “X-gram”. Therefore, it is generalizable among different buildings and naming conventions. We compare the proposed framework against commonly used baselines that include morphological analysis and widely used text mining techniques. Using two building cases from Singapore and two from the United States, we demonstrated that the framework outperformed baseline methods by 25.5%, with the measurement extraction F-measure of 87.2% and an average mapping accuracy of 91.4%.

Keywords

building management system (BMS), building energy modeling (BEM), auto-mapping, DBSCAN, metadata interpretation

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1 Introduction

With increasing number of sensors being deployed in buildings, the advent of the Internet of Things, and interoperability through Application Programming Interfaces, the operation data of buildings has become more easily available. With the collected data, there is an opportunity to extend building energy modeling (BEM) application throughout the building lifecycle, including retrofit analysis, Measurement and Verification, and operations and controls (Augenbroe 2004). However, the raw building management system (BMS) metadata is typically variant and irregular since different vendors have disparate customized rules to tag the points. For example, chilled water supply temperature can be tagged “SWT”, “Supply Temp”, “BMFWDTMP”,

etc. in different datasets. Similarly, different points such as cooling load and fan power can both be tagged with “KW” in the same dataset. Therefore, interpreting BMS metadata and mapping the data points to BEM input and output variables usually involves manual interpretation that requires comprehensive prior knowledge. This makes BMS metadata interpretation a labor-intensive and time-consuming process, which limits the application of BMS data in BEM.

It takes much time and extensive expert knowledge to build a high-quality building energy model. Accordingly, many studies were done using the information existing in Building Information Modeling (BIM) to facilitate this application of BEM (Gao et al. 2019). Welle et al. (2011) designed a middleware to transform the Industry Foundation Classes (IFC) architectural geometry into EnergyPlus thermal

E-mail: adrian.chong@nus.edu.sg

geometry for multidisciplinary design optimization. Ahn et al. (2014) developed and compared an automated and a semi-automated interface to convert IFC geometric information into EnergyPlus. Kim et al. (2016) created an IFC parser to translate the construction and material information from BIM to BEM. Chong et al. (2019) translated Green Building XML models into EnergyPlus models for continuous calibration. These methods were designed to match well-defined BIM schemas such as IFC, while BMS data rarely followed a standard format. Hence, a different approach is needed to map BMS points to BEM.

Similar to BMS point tags, the jargons in the field of bioinformatics, such as gene and protein names, have complicated synonyms, abbreviations, and are ambiguous to recognize. Therefore, it is likely to overlook them in the massive biometric text materials (Leser and Hakenberg 2005). The task of identifying and extracting unrecognizable names is known as Name Entity Recognition. Many recognition systems were built to retrieve these biometric words (Altschul et al. 1997; Gaizauskas et al. 2003; Kinoshita et al. 2005). However, these methods required substantial context information, which is often unavailable for BMS metadata.

Recently, with the increasing availability of data from the BMS, efforts were also attracted to facilitate its application. Bhattacharya et al. (2015a) designed a syntactic rule-based framework to interpret BMS point tags. Since the existing data schemas of building systems were not sufficient to model the BMS (Bhattacharya et al. 2015b), Balaji et al. (2018) proposed the Brick BMS data schema to describe the metadata of the BMS comprehensively. To promote the adoption of this schema, Koh et al. (2018) developed a framework based on transfer learning and active learning to normalize the original metadata of BMS datasets and map to the Brick schema. These works aimed at the same problem as our study and achieved acceptable accuracy in the case studies. However, both the rule-based and the transfer learning frameworks assumed that a certain level of similarity was reached among different buildings. Unfortunately, BMS point tags can be very dissimilar and follow different distribution in the feature space, affected by factors such as countries, building systems and naming conventions. For example, buildings in Singapore usually have only cooling systems, and a vendor may tag all points with abbreviations while another uses almost natural language. As a result, the frameworks might not be transferable, and improper transfer learning could lead to negative transfer that deteriorates prediction performance (Pan and Yang 2010).

In summary, manually interpreting and mapping BMS data points requires significant effort and domain knowledge. Meanwhile, the considerable variance and the lack of context information of raw BMS metadata limit the applicability of text mining methods designed for some

building modeling schemas or subjects from other fields. Aiming to promote the usefulness of BMS data, specific schemas were designed to model the BMS metadata and automated frameworks were proposed to facilitate mapping to the schemas. However, these methods based on supervised approaches expected a good similarity between the training and test data, which is hard to be met for BMS tags defined for different buildings and by different parties. Consequently, it is still challenging to automatically interpret BMS metadata in a scalable approach and to apply BMS data in actual BEM applications.

Accordingly, this study aims to design an automated and generalizable framework for BMS data point recognition and mapping to BEM. The objectives are to:

- 1) Develop the method based on unsupervised techniques to effectively facilitate the process, using EnergyPlus as the mapping subject.
- 2) Examine the performance and the generalizability of the proposed method using four case studies from different countries.

2 Methodology

Figure 1 illustrates an overview of the proposed BMS to BEM text mining framework, which consists of three main stages: (1) core information extraction, (2) tokenization and (3) fuzzy string matching. Density-Based Spatial Clustering of Applications with Noise (DBSCAN) (Ester et al. 1996) and a novel fuzzy string matching algorithm “X-gram” were applied to automate the BMS to BEM mapping process. EnergyPlus 8.8.0 input and output variables were used for the demonstration of the proposed framework. To bring clarity to the proposed framework, the mapping process of a typical BMS point, room temperature (as “RMTEMP” in the exemplary dataset and as “zone mean air temperature” in EnergyPlus), was used to exemplify the workflow.

2.1 Core information extraction

To facilitate the subsequent mapping, strings from the raw BMS metadata and EnergyPlus variables were first pre-processed to extract the core information and reduce the effects of noise on further mapping. In the datasets of BMS metadata, each sample contains a string, which usually follows the same format and naming rules within a building (defined by the same vendor or field engineer) but can be totally different from others. The EnergyPlus variables were extracted from the .eso and .mtr files because they have a stable format and contain spatial information about the buildings (EnergyPlus 2019). Table 1 shows examples of preprocessed BMS point tags and their corresponding variables in EnergyPlus.

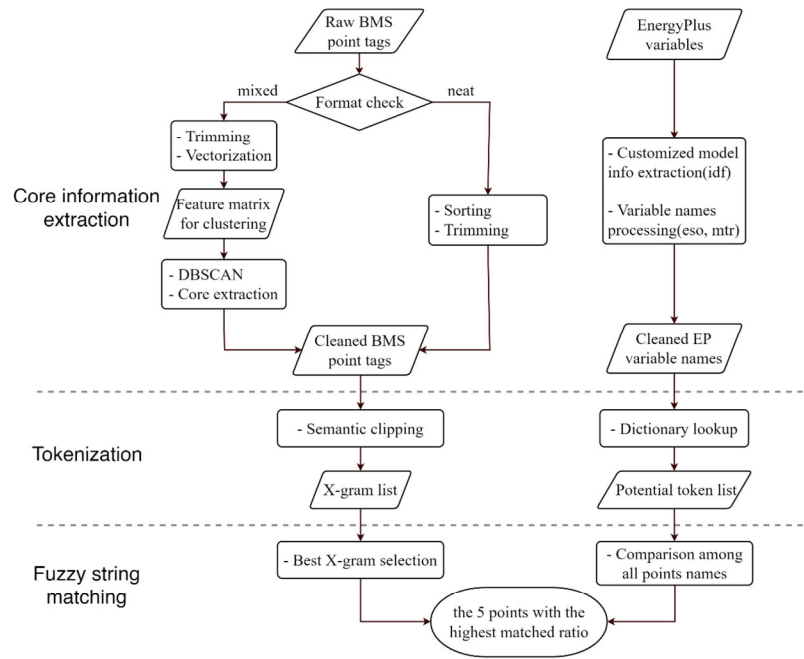


Fig. 1 Workflow of the text mining framework

Table 1 Preprocessed BMS points and corresponding EnergyPlus variables

BMS point tag	EnergyPlus variable	Source file
RMTEMP	Zone Mean Air Temperature	.eso
SASTATIC	Demand Side Inlet Pressure	.eso
SATEMP	Demand Side Inlet Temperature	.eso
PMKWHR	Electricity Cooling	.mtr
BMENGRY	Energy Transfer Cooling	.mtr

Both EnergyPlus input and output files were used to retrieve and preprocess the variables. The customized names and default measure type names were first extracted from the Input Data File (IDF). Then the extracted information was used to assist parsing the .eso and .mtr files and extract the measurement information. Meanwhile, field names of the relevant variables, such as “Watts per Zone Floor Area” and “Schedule Name”, in the IDF were taken as the EnergyPlus inputs. This step was generalizable among different energy models because energy models typically follow a consistent syntax and data model.

As for BMS metadata pre-processing, it was first checked whether the tags were segmented by one of the common delimiters such as dots and slashes. If the points were neatly tagged with a clear delimiter, the core part of the tag was extracted by simply trimming with the delimiter. Occasionally, the points needed to be sorted by delimiter first, but the whole dataset could still be preprocessed in batch with simple semantic rules. However, in other

cases, the core measurement information was mixed with location or other superfluous information in the string (e.g. “FV3_2_1RMTEMP”) and therefore semantic rules were not applicable to all the points. The framework took advantage that there were always sensors of the same type installed at different locations in buildings (e.g. room temperature sensor in different rooms), clustering the same type of points together and extracting the core measurement information from each cluster.

Trimming and vectorization were applied to prepare the dataset when clustering was needed. BMS point tags usually include a great part of meaningless strings such as “FACILITY” and “POINTVALUE” in the example, which is harmful for clustering and were trimmed off using the common delimiters. As in Fig. 2, the trimmed tags were transformed into a vector to quantify the similarity between each other. Similarity between each string (S_K) and all strings in the dataset (S_1 to S_N) were calculated using the Levenshtein Distances (Levenshtein 1965). The distance represents the similarity between strings by counting the minimum number of single-character edits (i.e. insertions, deletions, or substitutions) needed to change a string into another. With this approach, the dataset was transformed into a matrix, where each row featured a BMS data point.

Figure 3 visualized the location of BMS point tags in the feature space after vectorization. With points of the same type of measurement close to each other, DBSCAN was applied to separate the closer points from others. Unlike the popular clustering methods such as K -means, DBSCAN doesn’t require the pre-defined cluster number but defines clusters based on the density distribution (Ester et al. 1996).

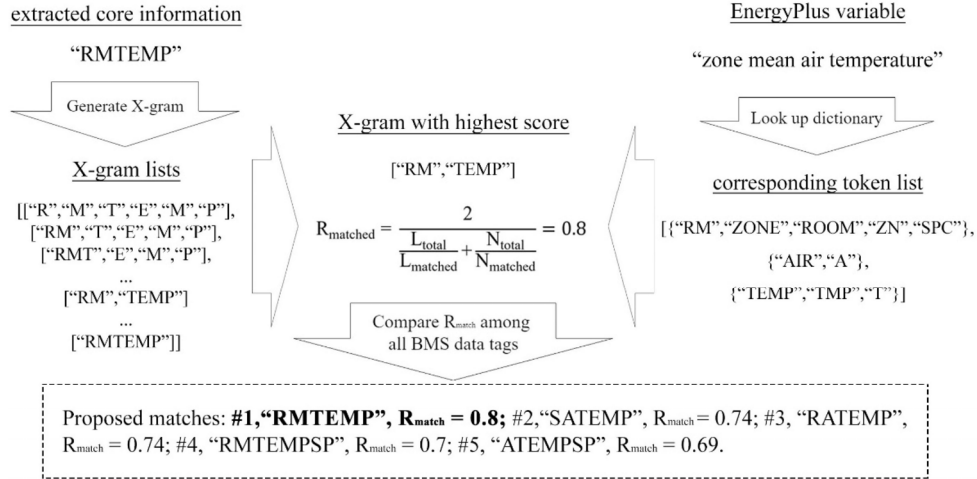


Fig. 4 An example of the X-gram based fuzzy string matching (right: EnergyPlus variable tokenization, left: BMS point tokenization, middle: similarity evaluation)

The tokenization of BMS point tags is to identify the meaningful substrings (e.g. "RM" and "TEMP" for "RMTEMP"). The tags usually contain words in irregular forms and abbreviations with different length, which makes the tokenization more difficult. The framework realized semantic clipping using the Algorithm 1, the intermediate results of which is illustrated on the left of Fig. 4.

In the algorithm, the cleaned BMS point tag S was first clipped in all possible ways to generate a set of X-gram segmentations. The substrings in each of the segmentation had variant length but aggregated to form the core string, which is why the method was named as "X-gram". Subsequently, the quality of each segmentation was evaluated against the list of token sets of the EnergyPlus variable by calculating the matched ratio defined as Eq. (1), where L_{total} is the length of the representative substring, L_{matched} is the

total length of matched substring, N_{total} is the total number of corresponding abbreviation sets, and N_{matched} is the number of matched sets. Eventually, the segmentation that got the highest score was kept to represent the BMS point.

$$R_{\text{matched}} = \frac{2}{\frac{L_{\text{total}}}{L_{\text{matched}}} + \frac{N_{\text{total}}}{N_{\text{matched}}}} \quad (1)$$

2.3 Fuzzy string matching

After tokenization, the fuzzy string matching was done by comparing the highest matched ratio R_{matched} of all the BMS points. The aim was to measure how well the X-gram segmentation of a point tag can be matched to the EnergyPlus variable token list. Since the absolute number of matches would cause issues when the point tag was prominently longer or shorter, the ratio (i.e. the number of matches over the total number) was used as the criterion. However, when calculating the matched ratio of BMS points, shorter tags could get a higher score more easily (e.g. "TMP", "STS"), while longer tags could hit more tokens when calculating the matched ratio of EnergyPlus variables (the longer the name was, the more likely single letters were mismatched, e.g. "SASTATIC"). The harmonic average of both sides was therefore adopted to eliminate the influence of unexpected mismatch.

Also, during the process of evaluation, to eliminate the effect of 1-gram mismatch and duplicative match (e.g. "S" and "T" in "SASTATIC"), the substrings in each X-gram segmentation were ranked by length and longer substrings were matched first. Matched tokens were marked and excluded for the further match so that a set of tokens was matched only once.

Algorithm 1: semantically clip BMS point tags by X-gram generation and selection

Inputs: cleaned BMS point tag S , potential token list L of the EnergyPlus variable

Outputs: selected X-gram segmentation SEG_{selected} , highest matched ratio

```

 $R_{\text{matched}}$ 
procedure X-gram( $S, L$ )
     $R_{\text{matched}} = 0$ 
     $n = \text{length}(S)$ 
    for  $i = 1$  to  $n$  do
         $SEG = [C_n^{n+1-i} \text{ ways to clip } S \text{ into } i \text{ segments}]$ 
        for item in  $SEG$  do
             $\text{score} = \text{calculate\_equation\_1}(\text{item}, L)$ 
            if  $\text{score} > R_{\text{matched}}$  then
                 $R_{\text{matched}} = \text{score}$ 
                 $SEG_{\text{selected}} = \text{item}$ 
            end if
        end for
    end for
    return  $SEG_{\text{selected}}, R_{\text{matched}}$ 
end procedure

```


In the end, to improve the robustness, BMS points with the top five highest R_{matched} were raised to map to the EnergyPlus variable. The entire fuzzy string matching process was illustrated in Fig. 4.

2.4 Evaluation metrics

The two most important parts of the framework, core information extraction and fuzzy string matching, were evaluated separately. To evaluate the results of clustering based core information extraction, the ground truth was generated by manually trimming the point tags. Compared with the ground truth, points with the core information correctly extracted were counted as True Positions (TP), those with wrong extractions were counted as False Positives (FP), and those classified as outliers were counted as False Negatives (FN). F-measure (Eq. (2)), defined as the harmonic mean of precision (3) and recall (4), was used to evaluate the performance of the core information extraction (Van Rijsbergen 1979).

$$\text{F-measure} = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \quad (2)$$

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \quad (3)$$

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \quad (4)$$

In comparison with the X-gram based fuzzy string matching, a baseline algorithm was also implemented. N-gram, a popular tokenization method (Hakenberg et al. 2005), was applied to represent the BMS point tags and Jaccard distance (Cheng et al. 2015) was used to quantify the similarity. To compare the mapping accuracy of the proposed and baseline methods, the ground truth was obtained by manually mapping the BMS points to their corresponding EnergyPlus variables. The mapping accuracy was defined as the number of successfully mapped variables divided by the total number of variables to map. If the 5 BMS points selected to map to the EnergyPlus variable included the correct one, the variable was counted as top-5 successfully mapped. If the BMS point with the highest score was exactly the correct one, the variable was counted as top-1 successfully mapped.

3 Results¹

3.1 Dataset overview

To demonstrate the performance and the generalizability,

the proposed framework was tested and analyzed in four building case studies. The overall description of the buildings is summarized in Table 2.

The proposed and baseline fuzzy string matching algorithms were tested in all the four buildings. Meanwhile, among the four building cases, BMS points in Buildings B, C, and D were neatly tagged, and the core information was perfectly extracted by a few semantic rules. Therefore, DBSCAN was only applied in Building A to extract the core according to the format check (see Fig. 1).

3.2 Core information extraction results

After trimming and vectorization, the raw metadata of Building A turned into an 815×815 matrix with each line as the feature vector of the corresponding point, which was then fed into DBSCAN (with parameters MinPt=2 and Eps=11). As partially shown in Fig. 5, DBSCAN generated 108 clusters and identified 7 points as outliers. The longest common substrings were extracted for each cluster as the representatives. Since there existed clusters with the same representative substring, the duplicated ones were removed. Finally, 62 types of points were fed into the tokenization and fuzzy string matching.

With the recall of 0.869 and the precision of 0.876, the F-measure of core information extraction was 0.872. All the desired point types were correctly extracted except that “SATEMPSP” (supply air temperature setpoint) was mixed up with “RATEMPSP” (return air temperature setpoint) and extracted as “ATEMPSP”, failing the final mapping of the corresponding EnergyPlus variable.

3.3 Fuzzy string matching result

According to the results, while both the X-gram and N-gram methods were able to identify all the meaningful substrings, N-gram also introduced a lot of meaningless noise and frequently took noises as correct matches by mistake. As in Fig. 6, the mapping result of “RMTEMP” exemplified the situation. Compared with the finally kept X-gram

Table 2 Overall description of the case study buildings

	Type	Location	System	Dataset size
A	Office	Singapore	Variable Air Volume cooling	815
B	Mixed-used	Singapore	Dedicated outdoor air cooling	2000
C	Mixed-used	Pittsburgh, PA	Radiant heating and cooling	267
D	Office	Pittsburgh, PA	VAV heating and cooling	3000

¹ Source code in Jupyter notebooks available at https://github.com/ideas-lab-nus/BMS_BEM_mapping

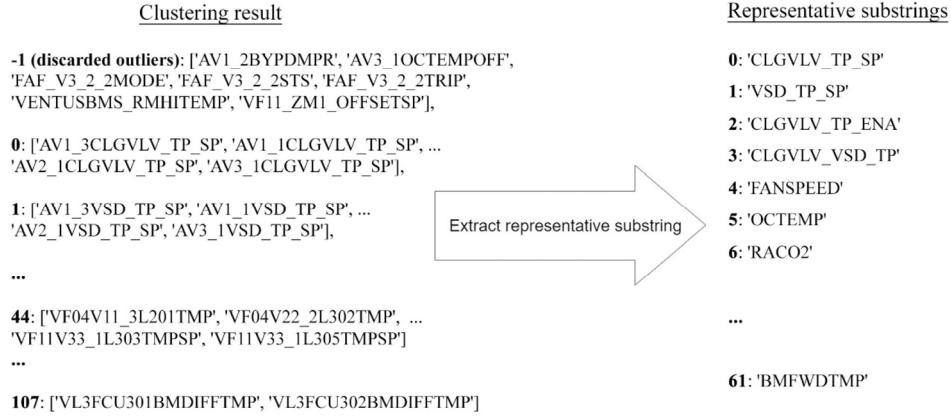


Fig. 5 DBSCAN based core information extraction results of Building A

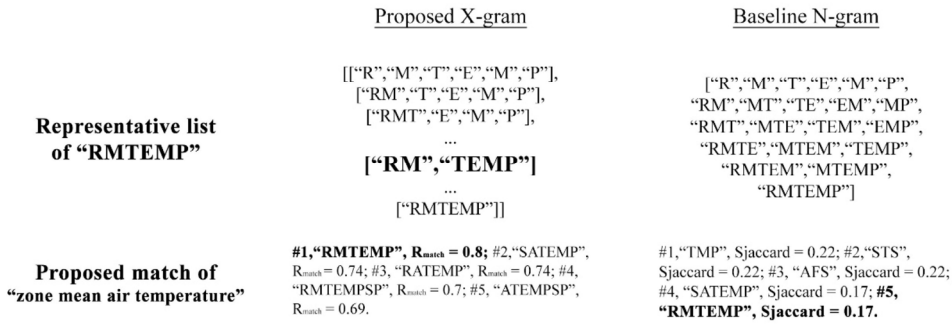


Fig. 6 An example of mapping result comparison

segmentation that only contained the correct substrings "RM" and "TEMP", the N-gram list included many unwanted substrings. Consequently, the score of "RMTEMP" was lowered and ended up in fifth place using N-gram. In the cases of other strings, the proposed X-gram showed more advantage, where the unwanted substrings given by N-gram caused the correct match to be outranked by incorrect points and not included in the top-5. For example, "OCTEMP", the abbreviation of "off coil air temperature", was the correct match of the EnergyPlus variable "cooling coil outlet temperature". Successfully identifying the meaningful substrings "OC" and "TEMP", X-gram found it as the best match from the dataset. Meanwhile, N-gram placed it in the ninth place, outranked by shorter tags with more "S"s and "T"s such as "STS" (status) and "TMPSP" (temperature setpoint).

Table 3 summarizes the top-5 and top-1 accuracies of the four building cases. As can be observed from Table 3, the proposed X-gram method outperformed the baseline in all the cases except getting the same top-5 accuracy in Building B. One thing to note is that there were many variables that did not have a correct match in the dataset but still got the recommended mapping results. Both methods could not avoid this problem because there was no apparent threshold for both criteria. Thus, these points were not counted when calculating the accuracy.

Table 3 The accuracy comparison against the baseline

	X-gram		N-gram	
	Top-5 acc.	Top-1 acc.	Top-5 acc.	Top-1 acc.
Building A	94.23%	51.92%	51.92%	25%
Building B	89.66%	48.28%	89.66%	37.93%
Building C	89.47%	73.68%	63.16%	47.36%
Building D	92.16%	52.94%	62.75%	35.29%

4 Discussion

4.1 Adaptability to various building cases

As discussed, different vendors define the BMS metadata in different ways when commissioning the system. For example, the four building cases tested in this study have completely different tagging conventions. The weak relationship between different buildings makes it hard to build a generalizable model. The proposed framework resolved this problem by (1) using unsupervised methods to extract the core information from the BMS point tags and (2) generating a universal dictionary to tokenize EnergyPlus input/output variables.

In the cases where the BMS points were not neatly tagged, it is difficult in the first place to figure out the

structure of the tags. More specifically, it is difficult to identify which part of the string contains the core measurement information. Given the condition that many sensors of the same type existed and that all points were tagged in a certain convention in the building, DBSCAN was demonstrated to be able to extract the core information with high recall and precision. However, note that this algorithm requires a certain level of consistency in the dataset. If points in one building were tagged by different vendors or randomly tagged, the performance of this clustering-based extraction would be degraded or even failed.

In the experiment, we generated a dictionary incorporating all words and abbreviations from the four test cases. The EnergyPlus variables were transformed into a list of tokens by looking up the dictionary. One example of this process is illustrated in Fig. 4. According to the testing result, with the subtle matching criterion, abbreviations from other datasets did not deteriorate the performance. This was because the matched ratio balanced out the influence of those unexpected mismatches. Thus, one single dictionary was applicable to different building cases as long as the synonyms and abbreviations were included.

4.2 X-gram based fuzzy string matching

The extracted core of BMS point tags are usually composed of abbreviations without a clear delimiter. The key to success interpretation is effectively identifying the meaningful tokens. By generating all possible X-gram segmentations and then filtering off the non-optimal ones, substrings with different length were analyzed together and the noise was greatly reduced. In this way, the finally picked segmentation only contained the meaningful substrings, while the traditional N-grams brought duplicated and unwanted substrings.

Compared with traditional criteria like Jaccard similarity, the proposed matched ratio counted the total length of matched substrings instead of the number of matched substrings, which takes full advantage of the X-gram segmentation and is more suitable for abbreviation recognition. However, for point tags with clear delimiters or without any abbreviation, the proposed method showed no superiority over the baseline. Many points in Building B were from third party sensors and the tags were almost natural language. For most of these points, both methods gave the correct match. X-gram failed on several points named with natural language, while N-gram did not recognize several point tags with abbreviations. This led to the same top-5 accuracy for both methods. Thus, the traditional N-gram actually worked better for full names. Since that abbreviations are more commonly used to tag BMS points,

the proposed X-gram achieved better performance in almost all test cases.

The other downside of the proposed X-gram based method was the longer runtime, especially for datasets with longer point tags. By iteratively clipping the string and examining the matched ratio, the computing steps grew cubically. Consequently, mapping a dataset including thousands of points took up to 15 minutes.

4.3 Accelerated mapping process

The framework was aimed to save human effort in the mapping process. However, like other data-driven methods, the framework could never reach 100% perfect results. With the diversified point names in the dataset, plus the many-to-many relationship between the full names and their abbreviations, it was expected that some incorrect points accidentally hit the tokens and got a close matched ratio as the correct match. Consequently, as shown in Table 3, the top-1 accuracy was always at a lower level. Thus, people still needed to check and refine the mapping result and it was impossible to completely avoid human interference in the mapping process.

Accordingly, we applied the concept of soft match to raise the top-5 mapping recommendations instead of an only proposed match. This involved minor further human interference to decide the final match, but picking from 5 with 90% confidence was much easier than checking the one with 50% confidence and going back to search in the hundreds. In the experiment, mapping with the framework took about one-fourth of the time used to generate the ground truth.

4.4 Directions for further development

Though the framework has been demonstrated to give promising mapping results, there is still work to be done to make it more reliable and scalable. Potential future development lies in the following directions:

- 1) Though the method was based on unsupervised methods and had no requirement on the similarity of a source dataset, the superiority against the supervised approaches was not demonstrated due to the lack of enough building cases. Being critical for the supervised methods, the required level of similarity between buildings was not quantified in the literature. It would be important and interesting to quantify the similarity and to investigate its influence on the performance of both types of methods.
- 2) In this study, the worst case was considered that only the BMS metadata were available as inputs. However, more

information such as the time series data and the units can be useful in actual applications. For example, Hong et al. (2015) extracted features from the time series data and used the features for sensors type classification. Considering that the availability of this complementary information is not guaranteed, it can be incorporated as an optional step to refine the mapping result with just the tags. For example, if the range of a temperature point were around 15 °C, it should not be recognized as room temperature.

- 3) The initiative of this study was to promote the use of BMS data for BEM. It is worth noting that the framework expects the BEM subjects to be named in a standard way. Since EnergyPlus is a very popular tool and has the stable schema-like format, its variables were taken as the mapping subjects. However, this temporarily stops the framework from being used for other simulation tools or other applications. Also, because EnergyPlus was not designed to model the BMS system, the overlap between its variables and BMS points is limited. Thus, as Koh et al. (2018) suggested, the mapping could be done towards a more well-organized and comprehensive data schema. This will improve both the performance and the scalability of the framework.

5 Conclusion

In this paper, we proposed and implemented a text mining framework to map the BMS points to EnergyPlus variables using just the metadata. The framework eased the mapping process by giving users the top-5 recommended matches out of hundreds or thousands of the raw point tags. Using 2 building cases from Singapore and 2 from the United States, we demonstrated that the framework reached the average accuracy of 91.4% and outperformed the baseline method by 25.5%. With the assistance of this framework, it is much easier for the energy modelers to find the corresponding BMS point for calibration and further application.

Making use of BMS data, BEM can be used for multiple applications beyond the buildings' design stage, such as retrofit analysis, Measurement and Verification, and operations and controls. This framework eases the requirement of prior knowledge to interact with BMS datasets, unchaining the application of BEM over the buildings' whole life cycle. Moreover, the framework eliminates the considerable effort spent to acquire and pre-process BMS data, which was usually underestimated in the literature. Thus, this study also contributes to BIM applications and data mining for buildings, increasing the interoperability of BMS data and reducing repetitive work among different parties.

References

- Ahn KU, Kim YJ, Park CS, Kim I, Lee K (2014). BIM interface for full vs. semi-automated building energy simulation. *Energy and Buildings*, 68: 671–678.
- Altschul SF, Madden TL, Schäffer AA, Zhang J, Zhang Z, Miller W, Lipman DJ (1997). Gapped BLAST and PSI-BLAST: a new generation of protein database search programs. *Nucleic Acids Research*, 25: 3389–3402.
- Augenbroe G (2004). Trends in building simulation. In: Malkawi A, Augenbroe G (eds), *Advanced Building Simulation*. London: Routledge. pp. 18–38.
- Balaji B, Bhattacharya A, Fierro G, Gao J, Gluck J, et al. (2018). Brick: Metadata schema for portable smart building applications. *Applied Energy*, 226: 1273–1292.
- Bhattacharya AA, Hong D, Culler D, Ortiz J, Whitehouse K, Wu E (2015a). Automated metadata construction to support portable building applications. In: *Proceedings of the 2nd ACM International Conference on Embedded Systems for Energy-Efficient Built Environments*.
- Bhattacharya A, Ploennigs J, Culler D (2015b). Analyzing metadata schemas for buildings: The good, the bad, and the ugly. In: *Proceedings of the 2nd ACM International Conference on Embedded Systems for Energy-Efficient Built Environments*.
- Cheng JC, Deng Y, Anumba C (2015). Mapping BIM schema and 3D GIS schema semi-automatically utilizing linguistic and text mining techniques. *Journal of Information Technology in Construction (ITcon)*, 20: 193–212.
- Chong A, Xu W, Chao S, Ngo NT (2019). Continuous-time Bayesian calibration of energy models using BIM and energy data. *Energy and Buildings*, 194: 177–190.
- EnergyPlus (2019). Available at <https://energyplus.net/documentation>
- Ester M, Kriegl HP, Sander J, Xu X (1996). A density-based algorithm for discovering clusters in large spatial databases with noise.
- Gaizauskas R, Demetriou G, Artymiuk PJ, Willett P (2003). Protein structures and information extraction from biological texts: the PASTA system. *Bioinformatics*, 19: 135–143.
- Gao H, Koch C, Wu Y (2019). Building information modelling based building energy modelling: A review. *Applied Energy*, 238: 320–343.
- Hakenberg J, Bickel S, Plake C, Brefeld U, Zahn H, Faulstich L, Leser U, Scheffer T (2005). Systematic feature evaluation for gene name recognition. *BMC Bioinformatics*, 6: S9.
- Hong D, Wang H, Ortiz J, Whitehouse K (2015). The building adapter: Towards quickly applying building analytics at scale. In: *Proceedings of the 2nd ACM International Conference on Embedded Systems for Energy-Efficient Built Environments*.
- Kim H, Shen Z, Kim I, Kim K, Stumpf A, Yu J (2016). BIM IFC information mapping to building energy analysis (BEA) model with manually extended material information. *Automation in Construction*, 68: 183–193.
- Kinoshita S, Cohen KB, Ogren PV, Hunter L (2005). BioCreAtIvE Task1A: Entity identification with a stochastic tagger. *BMC Bioinformatics*, 6: S4.

- Koh J, Balaji B, Sengupta D, McAuley J, Gupta R, Agarwal Y (2018). Scrabble: Transferrable semi-automated semantic metadata normalization using intermediate representation. In: Proceedings of the 5th Conference on Systems for Built Environments.
- Leser U, Hakenberg J (2005). What makes a gene name? Named entity recognition in the biomedical literature. *Briefings in Bioinformatics*, 6: 357–369.
- Levenshtein V (1965). Leveinshtein Distance.
- Pan SJ, Yang Q (2010). A survey on transfer learning. *IEEE Transactions on Knowledge and Data Engineering*, 22: 1345–1359.
- Van Rijsbergen CJ (1979). Information Retrieval. London: Butterworth-Heinemann.
- Welle B, Haymaker J, Rogers Z (2011). ThermalOpt: a methodology for automated BIM-based multidisciplinary thermal simulation for use in optimization environments. *Building Simulation*, 4: 293–313.